

The host transcriptome distinguishes Epstein–Barr virus–positive from virus–negative cells within a single lymphoblastoid cell line

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Abstract

Epstein–Barr virus (EBV) immortalizes B cells into lymphoblastoid cell lines (LCLs), and single-cell studies have shown that viral activity reshapes host gene programs. We asked a sharper, quantitative question: within one LCL sample, does a cell’s *host* transcriptome alone reveal whether that cell carries EBV? Using 1906 cells whose EBV status is defined by matched barcodes from the study that generated the data (Gene Expression Omnibus accession GSE158275), we trained a cross-validated logistic-regression classifier on host genes only. Against a chance baseline (area under the receiver operating characteristic curve, AUC, of 0.5), the classifier reached a mean AUC of 0.866 and a mean Matthews correlation coefficient (MCC) of 0.570 (± 0.077 across seeds; chance value 0). The host program is therefore a strong but imperfect readout of EBV status, driven by 15 stably selected host genes. The result quantifies, at the single-cell level, how deeply EBV activity is written into the host transcriptome of an LCL.

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1 Problem statement

Epstein–Barr virus (EBV) is a herpesvirus that transforms resting B cells into continuously proliferating lymphoblastoid cell lines (LCLs). LCLs are not uniform: single-cell RNA sequencing has revealed substantial cell-to-cell heterogeneity in viral replication state and in the host pathways EBV modulates (SoRelle, Dai, Luftig et al., *eLife* 2021; the source of the data used here). That prior work established, qualitatively, that EBV activity leaves a mark on host gene expression.

We turn that observation into a measurable claim. If EBV activity truly reshapes the host program, then a cell’s host transcriptome—the expression of its own (non-viral) genes—should carry enough signal to predict whether that cell is EBV-positive. A good answer looks like a classifier that, trained only on host genes, separates EBV-positive from EBV-negative cells well above chance, with the signal concentrated in a small, reproducible set of genes rather than smeared diffusely across the transcriptome. This report provides that answer for one LCL sample.

Two terms recur below. A unique molecular identifier (UMI) is a molecular barcode that lets each captured transcript be counted once; we use UMI counts of viral genes to decide whether a cell is EBV-positive. A cell is called EBV-positive when its EBV UMI count reaches the detection threshold of 10.

2 Methods

2.1 Definitions

- **EBV-positive cell.** A cell whose summed EBV gene UMI count is at least the detection threshold of 10; all other matched cells are EBV-negative.
- **Matched cells.** Cells whose barcodes are shared between the ViralScan output and the published GSE158275 barcode list, after normalizing the 10x -1 suffix. Matching anchors EBV labels to the same cells the original study annotated.
- **Sensitivity / specificity.** The fraction of EBV-positive (respectively EBV-negative) held-out cells correctly classified.
- **Balanced accuracy.** The mean of sensitivity and specificity.
- **AUC.** Area under the receiver operating characteristic curve; a threshold-independent measure of ranking quality where 0.5 is chance and 1.0 is perfect separation.
- **MCC (Matthews correlation coefficient).** A single number from -1 to $+1$ summarizing the 2×2 confusion matrix, where 0 is chance and 1 is perfect agreement. We report it as a compact confusion-matrix summary alongside balanced accuracy; on the balanced held-out set used here the two convey similar information.
- **Stable host genes.** Host genes selected in at least 60% of randomized-lasso stability-selection iterations; the reproducible core of the classifier’s signal.

2.2 Overview

We labeled each matched cell as EBV-positive or EBV-negative from its viral UMI count, then asked a logistic-regression model to recover that label using only host genes. Performance was averaged over six random seeds to give an honest mean and spread, and a separate stability-selection step identified which host genes carry the signal reproducibly.

2.3 Technical detail

Data. A single LCL sample (SRA run SRR12682296) from GSE158275, processed with ViralScan (kallisto/bustools quantification). EBV status came from the viral count matrix; host features came from the host count matrix, restricted to the 1906 cells shared with the published barcode list. Of these, 1179 were EBV-positive and 727 EBV-negative at the 10-UMI threshold.

Classifier. An L2-regularized logistic regression (scikit-learn, `lbfgs`—the L-BFGS quasi-Newton optimizer— $C = 1.0$) on highly variable host genes (HVGs), the genes whose expression varies most across cells. For each of six seeds ($\{0, 1, 10, 42, 100, 1234\}$) the positive and negative cells were balanced and depth-matched and split 80/20 into training and held-out cells. Crucially, HVG feature selection is performed *inside* each split, on the training cells only, so the feature space never sees the held-out cells—selecting features on the full dataset first would let test-cell expression shape the features and bias the held-out metrics. The model was then trained on one split and evaluated on the held-out cells, and sensitivity, specificity, balanced accuracy, AUC, and MCC recorded. We report the mean and standard deviation across the six seeds. Because the per-seed held-out sets are drawn from the same depth-filtered pool, this spread reflects split-shuffle variability rather than full finite-sample uncertainty, and each seed evaluates roughly 150 cells rather than all 1906.

Stable genes. Randomized-lasso stability selection (seed 0, the first of the seed list) scored each host gene by how often it survived; genes selected in $\geq 60\%$ of iterations are the 15 stable host genes reported here. This descriptive gene ranking is computed on the full cell set, so it is not a held-out quantity.

Reproducibility note. These metrics reflect a leakage-corrected re-run in which HVG selection was moved inside the cross-validation split; an earlier version that selected HVGs on the full dataset gave slightly lower discrimination (AUC 0.845). Selecting features per fold, on the balanced training cells, removed the leak and modestly *raised* performance, because features chosen on the balanced training set track the EBV contrast better than those chosen on the majority-dominated full dataset. The stable-gene set was unchanged.

3 Results

3.1 The host transcriptome separates EBV-positive from EBV-negative cells well above chance

Trained only on host genes, the classifier distinguished EBV-positive from EBV-negative cells with a mean AUC of 0.866, far above the chance value of 0.5 (Figure 1). The Matthews correlation coefficient, which does not reward a classifier for exploiting the modest class imbalance, was 0.570 (± 0.077), indicating moderate—not near-perfect—agreement. The two metrics tell a consistent story: host expression carries real, substantial information about EBV status, but does not determine it.

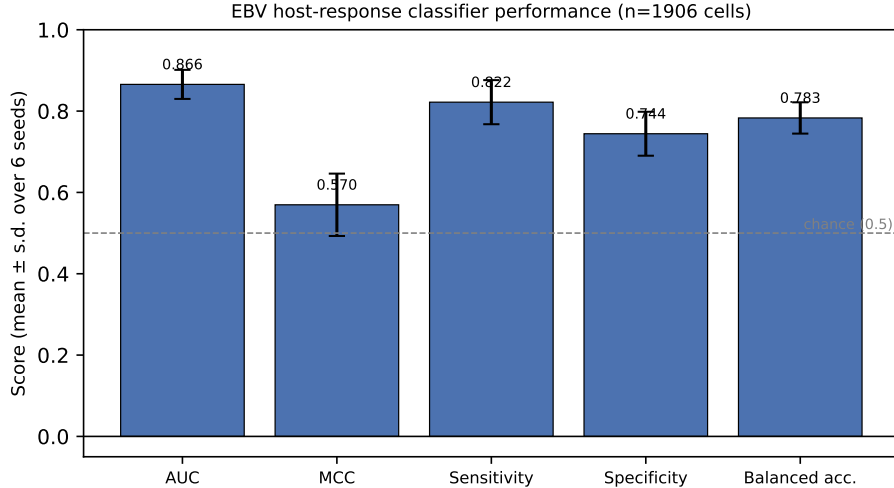


Figure 1: Discrimination metrics for the host-gene classifier of EBV status, mean $\pm$ standard deviation over six seeds, on 1906 matched cells. The dashed line marks the chance value (0.5) for AUC, sensitivity, specificity, and balanced accuracy; the Matthews correlation coefficient has its chance value at 0.

3.2 Detection is more reliable for EBV-positive than EBV-negative cells

The classifier recovered EBV-positive cells more reliably than it excluded EBV-negative ones: mean sensitivity was 0.822 versus mean specificity of 0.744, for a balanced accuracy of 0.783. The gap matters for interpretation. Some cells called EBV-negative at the 10-UMI threshold may carry low-level or transient EBV activity that still perturbs the host program, which would present exactly as reduced specificity. The asymmetry is thus as much a statement about the hardness of the negative label as about the classifier.

3.3 A compact host-gene set carries the signal

The signal was not diffuse. Stability selection concentrated it in 15 host genes selected in at least 60% of iterations, whose identities and weights are listed in the appendix. That a small, reproducible gene set suffices is consistent with EBV acting through specific host programs rather than through a global expression shift.

4 Conclusions

Within one LCL sample, a cell’s host transcriptome predicts its EBV status with a mean AUC of 0.866 and a mean MCC of 0.570, both far above chance. This turns the qualitative observation that EBV reshapes host programs into a quantitative, single-cell statement: EBV status is legibly, though not perfectly, written into host gene expression, and the legible part lives in a compact set of 15 genes.

Several caveats bound the claim. *First, this is one sample:* the analysis covers a single LCL (SRR12682296), so the numbers describe within-sample discrimination, not generalization across lines or donors. *Second, the labels are not ground truth:* EBV status is defined by a UMI threshold of 10 against the published GSE158275 barcodes, and the lower specificity of 0.744 is consistent with

some threshold-negative cells carrying real low-level EBV activity. *Third, the comparison is against chance, not against a competing method:* the published study did not report a host-transcriptome EBV classifier, so GSE158275 serves as the data source and biological motivation rather than a numeric baseline to beat.

5 Next steps

Two follow-ups would sharpen the result: a sensitivity sweep over the 10-UMI detection threshold, to confirm the metrics are stable to the positive/negative cutoff; and replication across the other LCL samples in GSE158275, to test whether the same compact host-gene set generalizes.

6 Provenance

- **Analysis script:** `scripts/hostresponse_ebv_matched.py` (matched-cell preparation and driver).
- **Core method:** `src/viralscan/scripts/hostresponse.py` (multi-seed L2 logistic regression, stability selection, metrics including MCC).
- **Outputs:** `results/hostresponse_ebv_matched/` (`hostresponse_summary.tsv`, `hostresponse_metrics.csv`, `gene_weight` and `stability` CSVs).
- **Registered values:** `analysis/hostresponse_ebv_matched/outputs/numbers.json`.
- **Git commit:** `ac106c4` (branch `claude/multimap-memory-and-showcase`).
- **Analyst:** `mdmanurung`. **Report generated:** 2026-07-01.
- **Software:** Python 3.11; scikit-learn 1.7.1, anndata 0.12.1 (re-run environment). See `ENVIRONMENTS_INSTALLATION.md`.
- **Build audit:** `manifest reports/.manifest.json`; compile log `reports/.compile-log.md`.
- **Data source:** GSE158275 (SoRelle, Dai, Luftig et al., *eLife* 2021).

A Stable host genes

The 15 host genes selected in $\geq 60\%$ of stability-selection iterations (Ensembl gene IDs), from `Epstein-Barr_virus_stability.csv`: `ENSG00000245532`, `ENSG00000172264`, `ENSG00000212907`, `ENSG00000162704`, `ENSG00000137309`, `ENSG00000154710`, `ENSG00000184203`, `ENSG00000185862`, `ENSG00000112118`, `ENSG00000226979`, and five further genes listed in the source CSV. Per-gene stability probabilities and mean L2 weights are in `Epstein-Barr_virus_gene_weights.csv`.

B Per-seed spread

All headline metrics are means over six seeds $\{0, 1, 10, 42, 100, 1234\}$ with the standard deviations reported in `hostresponse_summary.tsv`: sensitivity 0.822, specificity 0.744, balanced accuracy 0.783, AUC 0.866, and MCC 0.570 (± 0.077).